

Appendix 3

Reviewer Criteria

Principle	
Requirement	Ground Truth
The inlet velocity is 5 m/s. Change the nozzle outlet to make the output velocity 10 m/s	Outlet diameter 16.26 mm $\pm$ 10 %
The inlet velocity is 7 m/s, make the outlet velocity 20 m/s	Outlet diameter 13.61 mm $\pm$ 10 %
The inlet velocity is 7 m/s, make the throat velocity 12 m/s and the outlet velocity 10 m/s	Throat diameter 17.57 mm $\pm$ 10 %
The inlet velocity is 5 m/s. Draw the pressure at the following points: inlet, throat, and outlet	Implicit: ambient pressure = 1 bar
Calculate the volume of the nozzle. Print the equation and the result in the bottom right corner.	$V = \pi \times (11.5)^2 \times 92.74$ $V = \pi \times 132.25 \times 92.74$ $V = \pi \times 12,264.685$ $V \approx 38,528.2 \text{ mm}^3$ (or about 38.53 cm ³) + $V = \pi \times (4.425)^2 \times 43.8$ $V = \pi \times 19.5806 \times 43.8$ $V = \pi \times 857.63$ $V \approx 2,693.97 \text{ mm}^3$ $\approx 41,222 \text{ mm}^3$
The inlet fluid speed is 6.5 m/s. Calculate the liters per minute and draw it next to the inlet.	162.01 L/min
The flow rate is 6 L/min. Calculate the speed at the throat and draw it next to the throat.	2.15 m/s
The flow rate is 6 L/min. Calculate the pressure at the throat and draw it next to the throat.	At inlet pressure $p_{\text{atm}} = 101,325 \text{ Pa}$: throat pressure 99,043 Pa, i.e. 0.99 bar

Functional

Requirement	Ground Truth
My connecting flange has 10× screw holes and it uses M4 screws.	Changes hole count to 10, changes screws to M4, and changes the through-hole diameter to between 4.3 mm and 4.8 mm.
I want to mount it on a 75 mm flange with a 27 mm inlet connector.	Changes flange diameter to 75 mm and inlet diameter to 27 mm.
I want to use it as a high-pressure cleaner for light dirt. It is connected to my water tap.	Changes geometry to increase outlet pressure to 90–125 bar.
I want to use it as a high-pressure cleaner for medium dirt. It is connected to my water tap.	Changes geometry to increase outlet pressure to 125–170 bar.
I want to use it as a high-pressure cleaner for heavy dirt. It is connected to my water tap.	Changes geometry to increase outlet pressure to 170–220 bar.
I want to use it as a airbrush simulator. Can you make it feel the same as an original airbrush? It is connected to my water tap.	Changes geometry to increase outlet velocity.
Make the water cut 1 mm MDF board using a water tap.	Changes geometry to increase outlet velocity.
The screws are currently under too much stress and are failing. Make a design modification.	Changes flange or screws to hold more load.

Constructional

Requirement	Ground Truth
Set the flange diameter to 120 mm and the hole ring diameter to 80 mm.	Diameters get set.
Make the flange 15 mm thick.	Thickness changed.
Double the length of the nozzle.	Double the length of the nozzle from tip to flange.
Set the length of the nozzle to 132 mm.	Set the length of the nozzle from tip to flange.
Make the edges of the inlet smooth.	Inlet edges get rounded.
Remove the retaining groove.	Retaining groove removed.
Make the tip of the nozzle more steep.	Tip of the nozzle is more steep.
Move the flange 10 mm to the right.	Flange moved 10 mm.

Visual

Requirement	Ground Truth
Make the hatching red.	Hatching is red.
Increase the hatching density.	Hatching density is increased.
Remove the hatching.	Hatching gets removed.
Change the direction of the hatching.	Hatching direction changed.
Make the outer contour thicker.	Outer contour is thicker.
I can not read the dimensions, they look very small.	Dimension text gets larger.
This is a side view, add a front view.	A front view is added correctly.
Show the drawing four times in a grid with different color each.	Drawing is shown four times in different colors.